

1 Expanded View of Quality Data

Purpose

Use the function "expanded view of quality data", if you want to show further quality data in parallel to the input dialog. This is for example necessary if you want to permanently display a stamped drawing in addition to the inspection data. You can show assigned documents as a list or optionally in full screen mode.

Requirements

Meet the following requirements:

- Active license AIP-EQD with AIP 8.2.
- ctaip.exe as of version: 8.2.0.24
- caq_dc_t.dll as of version: 8.2.0.13
- caq72.dll as of version: 8.2.0.3

The planned machine or machine group must be included in the group "Workplace" in the MOC application *Inspection requirement*, on the level of the inspection step, tab *Inspection step*. This requires that the option *One inspection step for each inspection station* is set in the inspection planning. On the level of the characteristics, no inspection stations need to be assigned.

Activation

To activate the option, make the following entry in the file "caq_dc_t.ini". This file is stored in the AIP subfolder "functions".

```
[OPTIONS]
ACTIVATE_EQD=1
```

Features

Use the expanded view of quality data to show additional context-sensitive information in the CAQ recording of measured values. Possible contexts are: measured value, inspection point or characteristic. The application provides the following display elements to show the information:

- Control chart
- Histogram
- Data list
 - E.g. failures of a specific characteristic
- Document

view

E.g. image of a part

- Document list with document preview
Configure a list of documents with direct document view.
- Document list without document preview
Show a list and double-click a document that will then open in a separate window.

See the following layout examples:

Inspections for operation 0200 Deburring

Inspection point 27.07.2016 07:13:20

Shop floor workplace: DRM-001

Lot number: C347265

Batch: P23-45-12

Quantity: 0 PCS

Scrap: 0 PCS

Rework: 0 PCS

Usage decision:

- Description: other batch
- other material
- Rework
- rejection
- sales return Batch
- scrap

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Group/code: R2 01

Failure type:

Failure type	Date
scratch	20.01.2013
burr	21.01.2013
Xbar > upper action limit	24.05.2016
single value > upper tolerance limit	24.05.2016

Failure cause:

Failure cause	Date
transport damage	24.05.2016
wrong material	24.05.2016

Measure:

Measure	Date
Production on hold	24.05.2016
more severe inspection	24.05.2016

Inspection points

Showing measured values:

Inspections for operation 0200 Deburring

Characteristic Outer diameter

Inspection point 27.07.2016 07:13:20

Units to be checked: 1 * 1 PCS

Upper tolerance limit: 1.050 cm

Target value: 1.000 cm

Lower tolerance limit: 0.950 cm

Gage: Micrometer digital 0-25mm

Connection: offline

Meas. value: 1,000000 cm

Comment:

Inspector: 9999 Cavity: 1

The product is suitable to fix.

3D model of the part.

Recording of measured values

Configure the display elements (e.g. control chart, histogram and data list) using a configurable number of rows and columns of the recording of measured values. Configure every display element separately for each input type. Example:

Show a list of failures for a characteristic and show a control chart for the measured values.

Use XML files for the configuration. Change specific properties of an element directly in the respective tag of the XML configuration.

Sample configuration for a display element

The below example shows a document list with direct display of the contents of the selected data record.

```
<element class="DocumentControl" col="5" colspan="2" row="1" rowspan="4" description="document list with displaying a document">
  <settings>
    <workdir>documents</workdir>
    <context_change>characteristic</context_change>
    <document_source>grid</document_source>
    <header_visible>>false</header_visible>
    <view_align>vertical</view_align>
    <list_source>LIST;56</list_source>
    <list_dynamic_filter>RECTYP=%RECTYP%|BER=%BER%|PANNR=%PANNR%|PAUNR=%PAUNR%|AFO=%AFO%</list_dynamic_filter>
    <list_static_parameter></list_static_parameter>
    <columndef_section_name>QM-documents_EQD</columndef_section_name>
    <percentage_grid_usage>60</percentage_grid_usage>
    <can_zoom_document></can_zoom_document>
  </settings>
</element>
```

For further information refer to the document *Configuration_AIP-EQD*.

Integration of customer-specific components

In addition to the above-mentioned display elements and their configuration properties, you can also show customer-specific components in a specific display area of the AIP screen. To do so, the system sends the current node data to the component when you change nodes in the inspection list. Therefore, you can always refer to current CAQ data.

2 Alternative Inspection Start - Inspection without Operation Logon

Purpose

The following three fields of application have specific requirements regarding the inspection process. They are different to the usual in-production inspection on the production terminal.

- Incoming goods inspection
- In-production inspection at an exclusive inspection station (e.g. in the lab)
- Calibration

Use the Alternative Inspection Start for these three inspection processes. You do not need to log on an operation.

Requirements

Meet the following requirements:

- Active license AIP-EQD with AIP 8.2.
- ctaip.exe as of version: 8.2.0.33
- caq_dc_t.dll as of version: 8.2.0.15
- caq72.dll as of version: 8.2.0.7

Activation

The following fields of application have specific inspection modes:

- Incoming goods inspection
- In-production inspection at an exclusive inspection station (e.g. in the lab)
- Calibration

Only one of the modes can be active. Restart the terminal software if you change mode. For activation, one of the following configurations is required in the file „hytnrcfg.ini“ in the section „[CAQ->Optionen 0]“:

Inspection mode "Goods receipt inspection"

```
[CAQ->Options 0]
TERM_MODE= GOODS_RECEIPT
```

Inspection mode "Production inspection at an exclusive inspection station (e.g. in the lab)".

```
[CAQ->Options 0]
TERM_MODE=LAB
```

Inspection mode "Calibration"

[CAQ->Options 0]

TERM_MODE=CALIBRATION

The terminal uses the first entry found for "TERM_MODE".



If the parameter "TERM_MODE" in section "[CAQ->OPTIONS 0]" includes several entries, comment out the irrelevant ones.

Activate a specific main layout for the tile view of each of the inspection modes. The layout files can be found in the AIP subfolder "gui". Activate the main layout as follows:

Inspection mode "Goods receipt inspection"

Rename the file "I_main#lgoods#.xml" in "I_main.xml".

Inspection mode "Production inspection at an exclusive inspection station (e.g. in the lab)".

Rename the file "I_main#llab#.xml" in "I_main.xml".

Inspection mode "Calibration"

Rename the file "I_main#lcal#.xml" in "I_main.xml".

In addition, the CAQ option 1157 must be enabled and have the entry Y in the field *Value*. Also assign the parameter [GROUP:GEPLANT,MNR,MGRP,OPT_PLAN] in the field *Addition*. The parameter [GROUP:GEPLANT,MNR,MGRP,OPT_PLAN] groups the inspection step characteristics by assigning them to the fields "Planned", Machine group, "Machine" and "Inspection station". For each combination an inspection step is generated. The inspection steps will then include the contents of the grouping criteria in the respective fields. The inspection steps must include these grouping criteria. The system uses the grouping criteria to specify the respective inspection steps that are required for the three special inspection modes.

The system creates an inspection step, if option 1157 is configured accordingly for each combination of grouping criteria and if the option *One inspection step for each inspection station* is enabled in the inspection plan header.

For the inspection mode *Calibration*, you must set the parameter PAN_AU/AUNR_COPY in the field *Action* of the application *Area-Order type configuration* (Menu entry *System administration* → *System settings*). This configuration automatically generates a calibration inspection requirement if the calibration calendar includes a calibration order.

Logging of data

If logging is required, you must set the following parameter in the hytnrcfg.ini in section "[OPTIONS 0]".

TERM_MODE_DEBUG=ON



Using this parameter, the inspection data included in the spool directory is not deleted when the inspection dialog is closed. This is required in case of an upload with this inspection mode, for example.

It is best to set this parameter in the local file hytnrcfg.ini, to write-protect the file and to restart the AIP. After the secured logging (e.g. via upload), remove the write-protection, delete the parameter and restart the terminal.

Functionality – Inspection mode "Goods receipt inspection"

The inspection mode "Goods receipt inspection" is displayed in a special layout.

All possible inspection steps of this workplace are displayed. The displayed inspection steps are identified as follows:

1. The machine group of the inspection station is identified.
2. Using the specifications in the terminal configuration, the possible inspection steps are limited to the area type and the area.
3. The inspection steps of the machine group are identified, if the machine group matches the one of the inspection station. In addition, the inspection steps planned for the machine are identified, if the machine and the inspection station are identical.

As soon as the system generates a goods receipt inspection via interface in HYDRA, the inspection is displayed after a cyclic update.

You can configure the update interval in the file "caq72.ini" in the subfolder "packets" using the parameter "LOADCYCLE".

```
[DATACONTEXT_GOODS_RECEIPT]
LOADCYCLE=300
DATAPROVIDER_ID=PAUMNR
LIST=u_l_caq_inspstep_tnr
```

The standard layout can display up to 18 inspection steps at the same time. You can integrate the filter field and then filter by all displayed inspection step contents. By default, the displayed inspection steps are sorted in ascending order according to the date/time of inspection step generation. You can use the "caliper" button to access the inspection process.

The function to complete the inspection requirement is a special feature. This function automatically identifies inspection steps and inspection requirement results. If required, you can change them.

When you complete an inspection step, the system automatically sets the QM operation to "completed". Requirement: the QM operation must not be logged on and the processing code must define the OP as *inspection OP*. If the finished QM operation is the last operation to be finished in the order, the order is then set to the status "Completed".



If the inspection step is completed in the offline status, the operation is not automatically set to "completed". This would require a manual posting later on when the terminal is online again ("posting required"). If the required posting is not made, the inspection step is not completed. There is no message that the inspection step has not been completed.

Functionality – Inspection mode "In-production inspection at an exclusive inspection station"

This inspection mode does not require an operation logon either. Contrary to the in-production inspection at an exclusive inspection station requiring a QM logon, this inspection mode instantly shows all possible inspection points at this inspection station.

All possible inspection points of this workplace are displayed. The displayed inspection points are identified as follows:

1. The machine group of the inspection station is identified.
2. Using the specifications in the terminal configuration, the possible inspection steps are limited to the area type and the area.
3. The inspection steps of the machine group are identified, if the machine group matches the one of the inspection station. In addition, the inspection steps planned for the machine are identified, if the machine and the inspection station are identical.
4. For each of the identified inspection steps, all possible inspection points are identified.

As soon as an inspection point is generated when an inspection is due, the terminal shows the inspection point of the specific inspection station after the specified updating interval.

You can configure the update interval in the file "caq72.ini" in the subfolder "packets" using the parameter "LOADCYCLE".

```
[DATACONTEXT_LAB]
LOADCYCLE=300
DATAPROVIDER_ID=PPKTMNR
LIST=u_I_caq_inspoint_tnr
SECTION=DATACONTEXT_LAB_INSP_PT_MATURITY
```

In the list of the inspection point tiles, you can manually generate free inspection points using the "+" button. Using the button, the relevant inspection steps of this inspection station are identified and displayed in a list you can filter. Select an inspection point in the list and use the button "Generate inspection point". A free inspection point is generated for this inspection step. If you do not want to generate further inspection points, close the list of inspection steps manually.

Inspection steps can be completed in this inspection mode. Focusing on detail information on Workplace/Machine, you can complete an inspection step using the button "Complete". The relevant inspection steps of this inspection station are identified and displayed in a list you can filter (see above: generating an inspection point). Select an inspection step and use the button "Complete inspection step". When you complete an inspection step, the system automatically sets the QM operation to "completed". Requirement: the QM operation must not be logged on and the processing code must define the OP as *inspection OP*. If the finished QM operation was the last operation to be finished in the order, the order is then set to the status "Completed". If you do not want to generate further inspection points, close the list of inspection steps manually.



If the inspection step is completed in the offline status, the operation is not automatically set to "completed". This would require a manual posting later on while the terminal is online again ("posting required"). If the required posting is not made, the inspection step is not completed. There is no message that the inspection step has not been completed.

Functionality – Inspection mode "Calibration"

This inspection mode automatically produces a calibration inspection requirement including respective inspection steps when a calibration order is generated in the calibration calendar. If you want to generate a calibration inspection requirement via calibration order, the following conditions must be fulfilled:

- Configure the calibration inspection plan as follows:
 - Operation assignment: "One inspection plan for all operations"
 - IO + inspection station: "One inspection step (=IO) for each inspection station"
 - Generate new QM OPs: "None"
 - IO + generate characteristics: "When generating inspection requirements"

- The inspection plan characteristics for the calibration must be planned for a machine/machine group.

The inspection mode "Calibration" has a specific layout. The displayed inspection steps are identified as follows:

1. The machine group of the inspection station is identified.
2. Using the specifications in the terminal configuration, the possible inspection steps are limited to the area type and the area.
3. The inspection steps of the machine group are identified, if the machine group matches the one of the inspection station. In addition, the inspection steps planned for a machine are identified if the machine is the one of the current workplace.
4. You can configure the update interval in the file "caq72.ini" in the subfolder "packets" using the parameter "LOADCYCLE".

These inspection steps are instantly displayed in the inspection mode "Calibration" without operation logon.

```
[DATACONTEXT_CAL]
LOADCYCLE=300
DATAPROVIDER_ID=PAUMNR
LIST=u_I_caq_inspstep_tnr
```

A special feature of the inspection mode "Calibration" is that you can complete the calibration directly on the AIP when finishing the inspection step and requirement. In the process, you also select the test equipment status, the calibration result and the basis for calculating the next calibration.

If you complete the inspection requirement, the calibration order is automatically completed, too. As a condition, the respective QM operation must not be logged on and the OP must be defined as *inspection OP* via processing code.

The system resets the calibration in the calibration calendar in the same way as a maintenance activity.



If the inspection requirement is completed in the offline status, the operation is not automatically set to "completed". This would require a manual posting later on while the terminal is online again ("posting required"). If the required posting is not made, the inspection requirement is not completed. There is no message that the inspection requirement has not been completed.